

Smart Healthcare

Complete Project Description

Mobile App Development Reference Document

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1. Product Overview

Smart Healthcare is a digital health platform that connects patients with AI-assisted symptom analysis, virtual doctor-style chat, online pharmacy, clinic and hospital discovery with maps, appointment booking, and profile management. The product is branded **Smart Healthcare** in the user interface.

Tagline: Healthcare Made Simple - instant health insights, doctor consultations, medicine orders, and nearby clinics in one trusted platform.

2. Two Applications in One Repository

App	Path	Dev URL	Audience
Patient Portal	Root (app/, components/)	localhost:3000	Patients
Provider Portal	provider-portal/	localhost:3001	Staff (admin, doctor, clinic, pharmacy)

For a patient mobile app, focus on the main application and its API routes. A provider mobile app would be a separate product using the provider-portal APIs.

3. User Roles

Role	Purpose
patient	Default role; uses all patient-facing features
doctor	Provider portal - appointments, consultations
clinic	Clinic management operations
pharmacy	Inventory and order management
admin	Users, facilities, hospitals, stats, map administration

4. Patient App - Screens and Flows

4.1 Public (Unauthenticated)

Screen	Route	Behavior
Landing	/	Marketing, feature cards, live AI symptom demo (no login)
Login	/auth/login	Email and password authentication
Signup	/auth/signup	New user registration
Signup Success	/auth/signup-success	Confirmation page
Auth Error	/auth/error	Error display

4.2 Authenticated App Shell (/app/*)

Menu Item	Route	Purpose
Dashboard	/app	Home - stats, wellness chart, quick actions
Symptom Checker	/app/symptom-checker	AI symptom analysis
Doctor Chat	/app/doctor-chat	Streaming AI virtual doctor
Pharmacy	/app/pharmacy	Medicine catalog and shopping cart
My Orders	/app/orders	Order history and tracking
Find Clinics	/app/clinics	Facility list and interactive map
Appointments	/app/appointments	Book and manage appointments
My Profile	/app/profile	Personal and medical information

5. Feature Details (Mobile Implementation)

5.1 AI Symptom Checker

- API: POST /api/symptom-check
- Auth: Not required on landing demo; same endpoint when logged in
- AI Engine: Groq via Vercel AI SDK, streaming text response
- Request: symptoms, duration, severity (1-10), conversationHistory, isFollowUp
- Response: Streamed markdown-style sections (assessment, condition likelihood, severity, follow-ups)
- UX: Multi-turn conversation with follow-up questions
- Disclaimer: Educational only - not a medical diagnosis

5.2 Doctor Chat (AI Assistant)

- API: POST /api/chat (requires authentication)
- Request body: message, conversationId
- Response: Streamed text from Groq llama-3.3-70b-versatile
- Backend: Upserts consultations collection in MongoDB
- Note: HTTP streaming AI, not WebSocket or live human doctors

5.3 Pharmacy

- List medicines: GET /api/pharmacy/medicines (public)
- Checkout: POST /api/pharmacy/checkout (auth required)
- Medicine fields: name, description, price, stock, category, imageUrl
- Cart stored client-side in localStorage key smarthealthcare_cart
- Checkout sends items array and total; server decrements stock
- Categories: Cold and Flu, Pain Relief, Antibiotics, Vitamins, Digestive Health

5.4 Orders

- API: GET /api/user/orders (auth required)
- Returns: id (ORD-XXXXXX), order_date, total_amount, status, items
- Statuses: pending, confirmed, processing, shipped, delivered, cancelled

5.5 Find Clinics / Facilities

- API: GET /api/facilities (public)
- Data: Hospitals and clinics (seed data: Andhra Pradesh, India)
- Fields: name, facility_type, address, city, state, phone, rating, lat/lng
- UI: Search, filters, list view, Leaflet map integration
- Mobile: Use Google Maps, Mapbox, or react-native-maps

5.6 Appointments

- List: GET /api/appointments (auth)
- Create: POST /api/appointments with clinic_id, appointment_date, appointment_time, reason

- Status on create: pending (UI also shows confirmed, cancelled)

5.7 Profile and Health Data

- API: GET/PUT /api/user/profile (auth)
- Fields: email, first_name, last_name, phone, date_of_birth, medical_history
- Note: GET /api/health is a server ping only, not patient health records

6. Authentication (Critical for Mobile)

Current implementation uses MongoDB collections (users, sessions), bcrypt password hashing, and HTTP-only cookie sessionToken (UUID). Login returns id, email, role and sets a cookie. Current user via GET /api/auth/me.

Mobile auth options:

- Add Bearer token auth - return sessionToken in login response, accept Authorization header
- Use WebView with cookie jar (fragile on native apps)
- Migrate to Supabase Auth or Firebase and align backend

Important: README mentions Supabase, but patient auth in production paths uses MongoDB and cookies, not Supabase JWT.

7. Backend and Data Architecture

7.1 Technology Stack

Layer	Technology
Frontend	Next.js 16, React 19, TypeScript
Styling	Tailwind CSS 4, Radix UI, Framer Motion
API	Next.js Route Handlers (app/api/*)
Database (runtime)	MongoDB (MONGODB_URI, db: smarthealthcare)
AI	Vercel AI SDK + Groq (GROQ_API_KEY)
Reference schema	Supabase PostgreSQL in scripts/001_initial_schema.sql

7.2 MongoDB Collections (In Use)

Collection	Purpose
users	email, passwordHash, role, profile fields
sessions	token, userId, createdAt
appointments	User appointment bookings
medicines	Pharmacy product catalog
orders	Pharmacy order records
facilities	Hospitals and clinics
consultations	Chat session metadata

8. Complete API Reference (Patient App)

Base URL: NEXT_PUBLIC_APP_URL (e.g. http://localhost:3000)

Method	Endpoint	Auth	Description
POST	/api/auth/login	No	User login
POST	/api/auth/signup	No	User registration
POST	/api/auth/logout	Cookie	Logout
GET	/api/auth/me	Cookie	Current user
GET	/api/appointments	Yes	List appointments
POST	/api/appointments	Yes	Book appointment
POST	/api/symptom-check	No*	AI symptoms (stream)
POST	/api/chat	Yes	AI doctor chat (stream)
GET	/api/pharmacy/medicines	No	Medicine catalog
POST	/api/pharmacy/checkout	Yes	Place order
GET	/api/user/orders	Yes	Order history
GET	/api/facilities	No	Clinics and hospitals
GET	/api/user/profile	Yes	Get profile
PUT	/api/user/profile	Yes	Update profile
GET	/api/health	No	API health ping only

9. Provider Portal (Separate Mobile App Option)

- /login - Staff authentication
- /doctor - Appointments, messages, patient workspace
- /clinic - Clinic operations
- /pharmacy - Orders dashboard
- /admin - Stats, users, hospitals, facilities, map, settings

Admin API examples:

- GET /api/admin/stats - User counts, sessions, recent users
- GET/POST /api/admin/users, /api/admin/facilities, /api/admin/hospitals
- POST /api/admin/create-staff
- GET /api/pharmacy/orders, /api/pharmacy/dashboard

10. Environment Variables

- MONGODB_URI=your_mongodb_connection_string
- MONGODB_DB=smarthealthcare
- GROQ_API_KEY=your_groq_api_key
- GOOGLE_GENERATIVE_AI_API_KEY=optional
- NEXT_PUBLIC_APP_URL=http://localhost:3000

11. Suggested Mobile App Structure (MVP)

- 1. Onboarding - splash and value proposition
- 2. Auth - login, signup (forgot password not yet on web)
- 3. Home - quick actions mirroring /app dashboard
- 4. Symptom Checker - chat-style AI with streaming
- 5. Doctor Chat - authenticated AI assistant
- 6. Pharmacy - grid, search, filters, cart, checkout
- 7. Orders - list, detail, status tracking
- 8. Clinics - list, map, call and directions
- 9. Appointments - list and booking form
- 10. Profile - edit fields, logout, privacy links

Nice-to-have (not fully built on web):

- Push notifications (appointments, order shipped)
- Real human telemedicine (video/voice)
- Prescription upload
- Payment gateway integration
- Offline cart sync

- Biometric login

12. Mobile Technology Recommendations

Concern	Recommendation
Framework	React Native (Expo) or Flutter
Maps	react-native-maps / Google Maps SDK
Streaming AI	fetch with readable stream or SSE client
Auth	Implement Bearer token on backend first
State	React Query / TanStack Query
Cart	AsyncStorage + sync on checkout
UI	Match Smart Healthcare branding - primary blue, cards, lucide-style icons

13. Legal and Compliance

Landing page references Terms, Privacy, Cookie Policy, and HIPAA compliance (links are placeholders). AI features include disclaimers that output is not medical advice. For production, implement proper privacy policy, data retention, and regional compliance (HIPAA, GDPR, India DPDP, etc.).

14. Gaps Between README and Code

README Claim	Actual Behavior
Supabase auth	MongoDB + session cookies
WebSocket doctor chat	HTTP streaming AI (Groq)
Health records API	Profile medical_history field only
Google AI for symptoms	Groq is primary engine
PUT/DELETE appointments	Only GET + POST exist

15. Elevator Pitch

Smart Healthcare is an all-in-one patient platform: describe symptoms and get AI-powered analysis, chat with an AI health assistant, book clinic appointments, find hospitals on a map, order medicines from an online pharmacy, and manage your profile and orders - backed by a Next.js API and MongoDB, with a separate web portal for doctors, pharmacies, clinics, and administrators.